



The Perfect Answer
Revision Guide To...
Physics

LIE IGLSE

9-1 / A*-U

1st Edition

Hazel Lindsey, Dr. Caroline Gillespie

For use by Ashwin Shetty ashwinrangashetty@gmail.com ONLY. Not for redistribution.

Units and Symbols	3
1. General Physics	4
1.1 Length and Time	4
1.2 Motion	4
1.3 Mass and Weight	5
1.4 Density	5
1.5 Forces	6
1.6 Momentum	8
1.7 Energy, Work and Power	9
1.8 Pressure	13
2. Thermal Physics	15
2.1 Simple Kinetic Molecular Model of Matter.....	15
2.2 Thermal Properties and Temperature.....	16
2.3 Thermal Processes	20
3. Properties of Waves, Including Light and Sound	22
3.1 General Wave Properties	22
3.2 Light	23
3.3 Electromagnetic Spectrum	25
3.4 Sound.....	25
4. Electricity and Magnetism	27
4.1 Simple Phenomena of Magnetism	27
4.2 Electrical Quantities	28
4.3 Electric Circuits	31
4.4 Digital Electronics	33
4.5 Dangers of Electricity	35
4.6 Electromagnetic Effects	36
5. Atomic Physics	39
5.1 The Nuclear Atom	39
5.2 Radioactivity	40

Units and Symbols

Table courtesy of CIE IGCSE (9-1) Physics Syllabus 0972

Core			Supplement		
Quantity	Usual symbol	Usual unit	Quantity	Usual symbol	Usual unit
length	$l, h \dots$	km, m, cm, mm			
area	A	m^2, cm^2			
volume	V	m^3, cm^3			
weight	W	N			
mass	m, M	kg, g	mass	m, M	mg
time	t	h, min, s	time	t	ms
density	ρ	$\text{g}/\text{cm}^3, \text{kg}/\text{m}^3$			
speed	u, v	km/h, m/s, cm/s			
acceleration	a		acceleration	a	m/s^2
acceleration of free fall	g		acceleration of free fall	g	m/s^2
force	F	N			
gravitational field strength	g	N/kg			
			momentum	p	kg m/s
			impulse		Ns
moment of a force		Nm			
work done	W, E	J, kJ, MJ			
energy	E	J, kJ, MJ			
power	P	W, kW, MW			
pressure	p	N/m^2	pressure	p	Pa
atmospheric pressure		mm Hg			
temperature	θ, T	$^{\circ}\text{C}$			
			thermal capacity (heat capacity)	C	$\text{J}/^{\circ}\text{C}$
			specific heat capacity	c	$\text{J}/(\text{g } ^{\circ}\text{C}), \text{J}/(\text{kg } ^{\circ}\text{C})$
latent heat	L	J			
			specific latent heat	l	$\text{J}/\text{kg}, \text{J}/\text{g}$
frequency	f	Hz, kHz			
wavelength	λ	m, cm			
focal length	f	cm			
angle of incidence	i	degree ($^{\circ}$)			
angle of reflection, refraction	r	degree ($^{\circ}$)			
critical angle	c	degree ($^{\circ}$)			
			refractive index	n	
potential difference/voltage	V	V, mV			
current	I	A, mA			
e.m.f.	E	V			
resistance	R	Ω			
			charge	Q	C

NOTE: Standard black text is core content (grades 1-5). *Supplement content is in italics (grades 5-9)*

NB. Core content will be examined in papers 1 (MC) and 3. Supplement AND Core content will be examined in papers 2 (MC) and 4

1. General Physics

1.1 Length and Time

How is volume measured?

- Using a measuring cylinder
- Fill with liquid and read volume off scale

How is length measured?

- Use a rule for lengths from millimetres to a metre
- *A micrometer screw gauge measures very small distances*

How can an interval of time be measured?

- Clock, stop-clock or stopwatch
 - o Analogue with a hand moving around a circular scale
 - o Digital with numbers displayed on a screen

Calculate the average value of a short distance (e.g. the distance a javelin is thrown)

- Measure the distance multiple times (n times)
- Add all results together
- Divide the total by n

Calculate the average of a period of time (e.g. time it takes to run 100m)

- Measure the time multiple times (n times)
- Add the times together
- Divide the total by n

Calculate the period of a pendulum

- Measure the time taken for n swings
- Divide the time taken by n

1.2 Motion

Define speed

- Distance moved in a given timeframe
- Average speed = $\frac{\text{distance travelled}}{\text{time taken}}$

What is velocity?

- *The speed of something in a given direction of travel*
- *A vector (has direction and magnitude)*

Define acceleration

- *Change in velocity*

What is the formula for calculating acceleration?

- *Average acceleration = $\frac{\text{change in velocity}}{\text{time taken}}$*

How can distance travelled be measured from a speed/time graph?

- Calculate the area under the line

How do you find acceleration on a speed-time graph?

- Calculate gradient

What is deceleration?

- A decrease in speed
- Negative acceleration

What is free fall?

- Objects falling towards earth due to gravity
- Acceleration of free fall is measured in g
- The acceleration is constant

Describe the motion of objects falling without air resistance

- Downward acceleration of 9.8m/s^2
- Object continues to accelerate as no air resistance
- Until it reaches the bottom/ground

Describe the process of terminal velocity

- Weight acts downwards
- Drag acts upwards
- Object accelerates downwards
- Eventually weight = drag
- No resultant force, no acceleration, forces are balanced
- Terminal velocity (constant velocity) is reached

1.3 Mass and Weight

What is mass?

- A fixed measurement for any object
- Measured in grams (g) and kilograms (Kg)
- Property that 'resists' change in motion

What is weight?

- A gravitational force
- Measured in Newtons (N)
- Depends on strength of gravitational field
- The effect of a gravitational field on a mass

What is the equation for calculating weight?

- Weight = mass x gravitational field strength
- $W = m \times g$

How can weights (and hence masses) be compared?

- Using a balance

1.4 Density

How is density measured?

- Density is mass per volume: kg/m^3 or g/cm^3
- Calculated as $\text{density} = \text{mass} / \text{volume}$
- $\rho = m/V$

How can density of a liquid be measured?

- Measure the volume with a measuring cylinder or similar
- Measure the mass
- Use the formula $\rho = m / V$
- Remember to include units in answer

Example: 100l (0.1m³) of a liquid has mass 25kg, what is the density?

- Density = mass / volume
- Density = 25/0.1 = 250 kg/m³

How can the density of a solid with a regular shape be measured?

- Calculate the volume (length x width x depth)
- Measure the mass
- Use the formula $\rho = m / V$
- Remember to include units in answer

Example: a solid has mass 4 kg and its volume is 1m³, what is its density?

- Density = mass / volume
- Density = 4 / 1
- = 4 kg/m³

How can the density of an irregularly shaped solid be calculated?

- Using displacement to calculate the volume
- Measure the mass of the solid
- Use the formula $\rho = m / V$
- Remember to include units in answer

How to use given density data to predict if an object will float:

- Look at the relative density
- Relative density = density of a substance/density of water
- If relative density >1 an object will sink

1.5 Forces

Describe the effect of forces on a body

- A force applied to a body may cause a change in its size and shape

What is Hooke's law?

- *The extension of a spring is directly proportional to the force applied*
- *Provided its limit of proportionality is not exceeded*

Define elastic behaviour

- *The ability of a material to revert to its original shape after the forces causing deformation have been removed*

What happens if elastic limit is exceeded?

- *Material will no longer revert to original shape after the forces have been removed*

How you would plot and interpret an extension-load graph?

- A spring is stretched by a hanging load with the other end fixed
- A 100g mass will equate to a force of 1N
- Each additional 1N applied will produce the same extension again

Describe and explain the shape of an extension-load graph

- The line is straight and passes through the origin
- Load is proportional to extension
- Extension divided by load is always the same value
- *This value is called the spring constant (k)*
- *Load (F) = spring constant (k) x extension (x)*
- *The point this is no longer true is known as the 'limit of proportionality'*
- *Beyond this point the elastic limit has been exceeded, and the spring will not return to its original shape*

What is Newton's 1st Law?

- If forces acting on an object are balanced, the resultant force is zero
 - o Object at rest, stays stationary
 - o Object moving, continues to move in same direction and at same speed

What is Newton's 2nd Law?

- Acceleration is proportional to resultant force
- Inversely proportional to mass of object

How might an external force change the motion of a body?

- It may change its velocity (either it's direction or speed)

How to calculate resultant force acting along a line

- Add up all forces acting in useful direction
- Subtract all forces acting in the opposite direction

If the resultant force is zero, what change would you expect to see on the object?

- If the resultant force is zero → no overall force acting on the object
 - o If stationary it will remain stationary
 - o If moving it will maintain its velocity (direction and speed)

What is the relationship between force, mass and acceleration?

- $\text{Force} = \text{mass} \times \text{acceleration}$
- $F = m \times a$

E.g. If a ball is thrown with a force of 6N and has a mass of 3kg, what is its acceleration?

- $\text{Force (F)} = \text{Mass (m)} \times \text{Acceleration (a)}$
- So $a = F/m$
- $a = 6/3 = 2\text{m/s}^2$ (NOTE THE UNITS!)

Describe the force of friction

- Force between 2 surfaces that they exert against each other
- If one of those surfaces is the air then the friction is called 'air resistance'

What effects will friction have?

- Impede movement of the 2 surfaces past each other
- Generates heat if the surfaces are moving

Describe the forces that cause an object to move in a circle

- *When an object moves in a circle at a constant speed, its direction constantly changes*
- *Change in direction → change in velocity*
 - o *This is because velocity is a vector quantity (it has direction and magnitude)*
- *A change in velocity → change in acceleration*
- *Therefore an object moving in a circle is accelerating even though its speed is constant*

What is a moment?

- The turning effect of a force
- $\text{Moment} = \text{force} \times \text{perpendicular distance from pivot (fulcrum)}$

How would you increase a turning force (moment)? Give everyday examples

- Increase distance
- Increase force
 - o E.g. Using a longer spanner will increase the turning force on a nut
 - o E.g. Using a long lever to lift a heavy object the other side of the pivot (fulcrum)

Example: A crane lifts a beam weighing 1000N at a distance of 40m in front of it. What force is required on the counter balance 10m behind the crane to keep it in balance?

- Moment of the weight being lifted = 1000×40
- = 40,000Nm
- The same moment is needed the other side
- Moment = force $\times$ distance from pivot
- Force = moment / distance from pivot
- $F = 40,000\text{Nm}/10\text{m} = 4000\text{N}$

If the forces acting on an object are equal in all directions with no turning force what does this mean?

- There will be no resultant forces
- The system will be in equilibrium
- A plank in balance over its pivot point would show a system in equilibrium
- Clockwise moment = counter-clockwise moment

Define 'centre of mass'

- The point where the mass appears to be concentrated

How would you find the centre of mass of a plane lamina?

- Hang the lamina freely from a pin
- Centre of mass will hang directly under the pin
- Draw a line vertically from the pin, the centre of mass is along this line
- Repeat the above steps from a different edge
- Where the 2 lines intersect is the centre of mass

What is the relationship between the centre of mass and an object's stability?

- If centre of mass is over the base an object will be stable
- If centre of mass passes outside of the base an object will be unstable
- Objects with a low centre of mass and wide base are more stable

What two qualities must a force have to make it a vector?

- Magnitude
- Direction

How are vectors different to scalars?

- Scalars have only a magnitude, not a direction (e.g. mass)

How would you use a graphical method to find the resultant of two vectors?

- Draw 2 lines from an origin point 'O' to represent each vector
- The angle between the lines must be accurate
- The lines must be in scale to the force, e.g. 1mm per Newton
- Complete the other 2 sides of the parallelogram
- The length and direction of the line joining 'O' to the opposite corner gives the resultant vector

1.6 Momentum

What is momentum?

- It is a measure of the energy required to stop an object moving
- An object has momentum if it has mass and is moving
- Momentum is always conserved
 - o Momentum before = momentum after

What is the equation for momentum?

- Momentum = mass $\times$ velocity
- $p = m \times v$

What is the unit of momentum?

- kg m/s

Example: A railway carriage weighing 10,000kg is travelling at 7m/s when it bumps into a second carriage that is stationary and weighs 9000kg. They move away together. What is the velocity of the carriages together? Give your answer to 3 significant figures.

- Momentum (p) = mass (m) x velocity (v)
- Momentum before = momentum after
- Momentum BEFORE collision:
 - $10,000\text{kg} \times 7\text{m/s} = 70,000\text{kg m/s}$
- Therefore momentum after the collision = 70,000kg m/s
- Mass AFTER the collision:
 - $10,000\text{kg} + 9000\text{kg} = 19,000\text{kg}$
- $p = mv$ can be re-arranged to $v = p/m$
 - $v = 70,000 / 19,000$
 - $v = 3.68\text{m/s}$

What is impulse?

- A measure of the change in momentum
 - Change in momentum and time over which it is applied
- Usually measured in newtons, which will have the same value as kg m/s

Example: A simulated car crash has a safety dummy in the drivers seat that weighs 70kg. When crashed into a wall the dummy is travelling in the car at 15m/s and comes to rest after the crash. What is the impulse applied to the crash dummy?

- Impulse = change in momentum
 - As an equation $Ft = mv - mu$ (v is velocity at the end, u is velocity initially)
- So:
 - $Ft = 70 \times 0 - 70 \times 15$
 - $Ft = 0 - 1050$
 - $Ft = 1050\text{N}$
- Force = $\frac{\text{change in momentum}}{\text{Time}}$
- $F = \frac{(mv - mu)}{t}$

Explain how seat belts/crumple zones/air bags prevent serious injury

- Same momentum change
- But time of impact increases
- Reducing force felt
- Seat belt stretches increasing area over which force acts
- Pressure on body reduces

1.7 Energy, Work and Power

Outline the types of energy that an object may possess

- Kinetic
- Gravitational potential
- Elastic (strain)
- Chemical
- Electrical
- Nuclear
- Internal (thermal)
 - Potential energy = energy an object has due to a change in position, shape or state

How might an event or process change the energy an object has?

- Energy can never be destroyed, just transformed or transferred
- A process or event may transform energy
 - E.g. burning fuel will transform chemical potential energy into thermal energy
- A process or event may transfer energy
 - E.g. a snooker ball striking another will transfer its kinetic energy

What is a Joule?

- The work done when a force of 1 Newton moves an object 1m in the direction of the force

What is the equation for kinetic energy?

- $KE = \frac{1}{2}mv^2$
- Kinetic energy = $\frac{1}{2} \times \text{mass} \times \text{velocity}^2$

Example: A snooker ball weighing 0.2kg is moving at 2m/s when it strikes 2 other identical balls at rest. The initial ball stops still while the others move away. How much kinetic energy has been transferred in total to the 2 moving balls?

- $KE = \frac{1}{2}mv^2$
- $KE = \frac{1}{2} \times 0.2 \times 2^2$
- $KE = 0.4J$

Give an example of energy transfer by electrical working

- A battery has chemical potential energy
- An electrical current can transfer this energy into a resistor
- The electrical energy transferred to the resistor will transfer thermal energy

Give an example of energy transfer by mechanical working

- Forces transferring energy
- E.g. applying force to a lever will transfer an equal moment on the other end

What is the equation of gravitational potential energy?

- Gravitational potential energy = mass(m) x gravity(g) x change in height (Δh)

Example: A 1kg rock is held 2m above the end of a lever. How much kinetic energy is transferred to the object resting on the other end? Assume the system is 100% efficient.

- Initial potential energy of the rock = final kinetic energy of the object
- Gravitational potential energy = mass(m) x gravity(g) x change in height (Δh)
- GPE of the rock = $1\text{kg} \times 10\text{N/kg} \times 2\text{m}$
- Energy = 20J
- Because it is 100% efficient the kinetic energy at the end is 20J

Give an example of energy transfer by heating

- The sun has nuclear and chemical potential energy
- This energy is transferred into the surrounding solar system through heating

Give an example of energy transfer by waves

- E.g. Sound
- Speaking transfers kinetic energy from the vocal cords to the air
- Sound waves transfer this energy to the surrounding area

Explain why, in any event or process, all the energy at the start doesn't end up in a single form at the end.

- Energy will dissipate (spread out) across all the objects and into the surroundings

Describe how chemical potential energy in fuel can generate useful electricity

- Fuel (e.g. oil, coal, gas) is burned, transferring chemical potential energy to thermal energy
- Thermal energy used to turn water to steam which is kept under high pressure
- This pressurised steam turns turbines linked to generators
- The generators produce electricity

What are the advantages and disadvantages of generating power from fossil fuels?

Advantages	Disadvantages
• Large amounts of energy in small amounts of fuel • Reliable and predictable energy delivered • Cheap to set up relative to other methods	• Large amounts of pollution; CO₂ contributes to global warming, SO₂ leads to acid rain • Not renewable, it will run out one day

How can water be used to produce electricity?

- Waves: the energy of waves could be used to drive a generator
 - Few, if any, devices have proven able to harness waves' vertical motion
- Hydroelectric: storing water behind a dam and releasing it to flow past a turbine and drive a generator
- Tidal: incoming tide captured behind a dam and released past turbines to drive a generator

What are the advantages and disadvantages of water based energy sources?

Wave energy - Advantages	Disadvantages
• Renewable • Easily accessible • Very low environmental impact	• Technically very difficult • Energy unreliable • Depends on the wave state any given moment

Tidal energy - Advantages	Disadvantages
• Renewable • Reliable and predictable • No pollution produced	• Very expensive to build • Large scale local impact on the coast (including flooding) • Few areas are suitable

Hydroelectric power - Advantages	Disadvantages
• Reliable and renewable energy • No pollution • Free power source	• Expensive to build dams • Large impact of flooding and damage • Potential catastrophe if dam fails

How is it possible to harness energy direct from the sun to produce electricity?

- Solar power
- Dark coloured solar panels heat water to drive electricity generation
- Solar cells made from material that produces electric current when absorbing light energy

What are the advantages and disadvantages of generating power this way?

Advantages	Disadvantages
• Renewable • No pollution • Solar panels are relatively cheap	• Unreliable sunshine • Huge areas needed to generate enough power (10m² to power a kettle!)

How could you provide electricity from wind power?

- Wind turns a large turbine that drives a generator

What are the advantages and disadvantages of wind power?

Advantages	Disadvantages
• Renewable • No pollution • Cheap (after initial cost of turbines)	• Large areas of turbines need to be built in remote pieces of the environment • Unreliable amounts of wind

How can large amounts of geothermal energy generate electricity?

- Water piped underground to be heated by geothermal energy
- Rises as pressurized steam to drive a generator

What advantages and disadvantages does geothermal energy have?

Advantages	Disadvantages
• Low environmental impact • No pollution • Reliably available in the right areas	• Only some areas are suitable • Drilling deep enough is technically difficult • Expensive

Describe the process of producing power from nuclear fuels.

- Most nuclear fuels contain uranium
- The process of splitting uranium nuclei is called NUCLEAR FISSION
- The heat produced by fission reaction is used to heat water to steam
- The steam powers generators

How is nuclear power different from the way the sun produces energy?

- *The sun:*
 - Nuclei of atoms are combined together
 - This is called NUCLEAR FUSION
- *Nuclear power:*
 - Nuclei of atoms split in a controlled reaction
 - This is called NUCLEAR FISSION

What are the main advantages and disadvantages of power from nuclear fission?

Advantages	Disadvantages
• Reliable power generation • Large amounts of energy for small amounts of nuclear fuel • No air pollution generated	• Expensive power stations with high safety standards are needed • Nuclear waste is highly toxic and stay radioactive for 1000s of years

Which energy sources DO NOT rely on the sun?

- *Energy from the sun does or has produced most of energy on the planet*
- *The only exceptions are:*
 - *Geothermal*
 - *Nuclear*
 - *Tidal (produced from the gravitation pull of the moon)*

What is meant by efficiency?

- Proportion of useful energy or power produced compared to the energy or power put in
- Expressed as a percentage

How would you calculate efficiency?

- $\text{Efficiency} = \frac{\text{useful energy output}}{\text{total energy input}} \times 100$
- Or $\text{Efficiency} = \frac{\text{useful power output}}{\text{total power input}} \times 100$

What is meant (in scientific terms) by 'work done'?

- Work is a measure of a force moving an object
- Whenever work is done, energy is transferred
- Work done = energy transferred
 - o E.g. Potential energy might do work sliding a brick across the ground
 - o Due to friction this kinetic energy dissipates as sound and heat
 - o The loss of energy means the brick stops moving
 - o Potential energy is transformed to kinetic energy, which is transformed to sound and thermal energy

How would you calculate work done?

- $\text{Work done (W)} = \text{force (F)} \times \text{distance moved (d)}$
- Work done is also equivalent to the change in energy
- $W = Fd = \Delta E$

What is the relationship between power and work done?

- Power is a measure of how quickly work is done
 - o E.g. 2 different sized engines:
 - A small engine applies 5N to an object and moves it 5m in 30s
 - A large engine applies 5N to an object to move it 5m in just 10 seconds
 - Both engines do the same work
 - The larger engine has more power
 - o E.g. light bulbs
 - A more powerful light transforms energy into light more quickly so will be brighter than a less powerful light

How would you calculate power?

- $\text{Power (P)} = \frac{\text{change in energy}}{\text{time taken}}$

Example: A crane lifts 1000N block from the ground 20m in the air, taking 20 seconds to do it.

What power does the crane have?

- o $\Delta E = W = Fd$
- o $\Delta E = 1000 \times 20 = 20,000\text{J}$
- o $P = \frac{20,000\text{J}}{20\text{s}} = 1000 \text{ watts}$

1.8 Pressure

What is the equation for pressure?

- $\text{pressure} = \frac{\text{Force}}{\text{Area}}$
- $p = F/A$

How would you calculate the pressure an object applies to the ground?

- A 1N force applied to an area of 1m = 1N/m²
- = 1 pascal (Pa)

What is the relationship between force, area and pressure?

- A force applied at right angles to a surface will be spread evenly across it
- The same force on a small area will have higher pressure
- The same force applied to a larger area will have lower pressure

Give examples of this relationship

- Studs in a football boot sinking into the ground while a trainer does not
- Wall foundations giving a wide base to stop the wall sinking into the ground
- A drawing pin with a wide top but very thin tip applies enough pressure to be pushed into a wooden board

How does a mercury barometer measure atmospheric pressure?

- Atmospheric pressure acts on a reservoir of mercury
- Pushes the mercury up a sealed tube containing a vacuum
- Normal atmospheric pressure will support a column of mercury 760mm high
- Small pressure changes cause level of mercury in the tube to rise or fall

How does a manometer work?

- Measures pressure difference
- A simple manometer —> U shaped tube with mercury in it
- With no pressure difference the mercury will settle with an even height each side
- Increase in pressure on one side will force the mercury higher on the other side
- Difference in height is the difference in pressure between each side of the tube in mmHg

Describe how the pressure a liquid exerts is related to the density of the liquid and the depth below the surface that the pressure is measured

- The pressure a liquid exerts increases with depth as the weight of liquid above increase
- The more dense the liquid the more pressure it will exert at a given depth
- The pressure of the liquid acts in all directions
- In an open container pressure is related only to depth, not shape of container

How would you calculate the pressure a liquid applies at any given depth?

- $p = h\rho g$
- p = pressure; h = depth; g = gravity; ρ = density

Give examples of practical implications where pressure increases with depth

- Dams have thicker walls at the bottom to handle the higher pressure
- Deep sea submarines have to be built to withstand very high pressures

2. Thermal Physics

2.1 Simple Kinetic Molecular Model of Matter

What are the properties of a solid?

- Fixed shape
- Fixed volume

Describe the molecular structure of a solid

- Molecules held in a rigid structure
- Molecules held close together
- Molecules vibrate around a fixed point - cannot move past each other
- *This arrangement of molecules is why solids can not change shape or volume*

What are the properties of a liquid?

- Flexible shape
- Fixed volume

Describe the molecular structure of a liquid

- Molecules are close to each other and attract each other
- Molecules vibrate so much that the attraction does not fix them in position
- Molecules can move past each other although they are held close together
- *This is why liquids can flow and change shape but not change volume*

What are the properties of a gas?

- Flexible shape
- Changeable volume
- Will rapidly spread to fill the space it is in

Describe the molecular structure of a gas

- Molecules are almost totally free of attraction to each other
- Move freely at high speed colliding with each other and the edge of the container they're in
- *A lack of attraction and movement is why a gas changes shape and volume*

How does a change in temperature effect the movement of molecules in a gas?

- Increased temperature = molecules move faster with more energy
- Decreased temperature = molecules move slower with less energy

If a gas is heated in a container with a fixed volume, why will the pressure rise?

- *Temperature rises → particles move with more kinetic energy*
- *More collisions with container walls*
- *Increased collisions against the container walls = increased pressure*

What is kinetic theory?

- Used to explain the properties of solids, liquids and gases
- All matter is made up of minute particles
- Particles are constantly moving
- Particles attract each other, with weaker attraction when further apart

What is Brownian Motion?

- Particles in suspension move about randomly

How is this evidence of the kinetic theory?

- Newton's first law means that particles should keep a constant velocity, not move randomly
- A force must therefore be acting on particles to make them change direction
- This would be predicted by kinetic theory as:
 - The molecules a particle is suspended in constantly bombard it
 - *Small, fast moving molecules can move the large suspended particles*

Describe the process of evaporation

- A liquid slowly changing into gas (below its boiling point)
- Some of the liquid's molecules have more energy than others
- Some molecules have enough energy to escape the surface of the liquid and become gas

How will temperature, surface area and moving air across the surface affect evaporation?

- *Temperature:*
 - *Higher temperature means more molecules have high energy*
 - *More molecules escape → more evaporation*
- *Surface area:*
 - *Higher proportion of molecules near the surface*
 - *More molecules can escape → more evaporation*
- *Moving air:*
 - *Escaped molecules are moved away from liquid*
 - *Prevents molecules returning to the liquid → faster evaporation*

Why does evaporation cool the liquid?

- Kinetic theory states that high temperatures = particles with high energy
- Evaporation removes highest energy particles
- Particles left behind have low energy = low temperature

Why does an object in contact with an evaporating liquid cool down?

- *Thermal energy is transferred from the object to the liquid molecules*
- *Molecules retain that energy when they escape the liquid*
- *Leaves a cooler liquid to which more of the object's thermal energy is transferred*

How will changing temperature affect the pressure of a fixed volume of gas?

- Higher temperature → higher kinetic energy → increased pressure
- Lower temperature → less kinetic energy → lower pressure
- Pressure is directly proportional to the temperature (in kelvin)

How will changing volume affect the pressure of a gas at a constant temperature?

- Governed by Boyle's Law
- Pressure is inversely proportional to volume (if temperature is constant)
- If volume halves, pressure doubles
- If volume doubles, pressure halves

How can Boyle's Law be expressed as an equation?

- *The pressure multiplied by the volume is always the same*
- $pV = \text{constant}$
- $p_1 \times V_1 = p_2 \times V_2$

E.g. If a gas has a volume of 3cm^3 and a pressure of 4 atmospheres and the pressure is lowered to 1 atmosphere, what will be the new volume? Assume temperature stays the same.

- *Because $pV = \text{constant}$ then $p_1 \times V_1 = p_2 \times V_2$*
- *So $4 \times 3 = 1 \times V_2$*
- *So $V_2 = 12\text{cm}^3$*

2.2 Thermal Properties and Temperature

Why do solids and liquids expand when they're heated?

- At higher temperatures particles have more energy → move more
- In solids and liquids this means particles vibrate more
- Increased vibration results in expansion to accommodate it

Give examples of how expansion affects us

- Thermometers: the liquid inside expands and contracts
- Concrete is reinforced with steel as both materials expand the same amount
- Power cables are left slack to allow for contraction on cold days
- Bridges have small gaps to allow for expansion of road surface on hot days

Explain why solids expand less than liquids or gases when heated

- Solids have tight arrangement with strong attraction between molecules
- This limits expansion
- Liquids and gases have weaker attractions
- Same energy increase therefore leads to greater expansion

Explain how liquid-in-gas and thermistor thermometers work

Liquid-in-gas thermometers	Thermistor thermometers
• Volume of liquid changes with temperature • Volume and therefore temperatures can be read from a scale	• Electrical resistance of a metal varies with temperature • Thermistor allows a higher current to flow with higher temperature • Higher current can be detected and displayed on the screen

How are thermometer scales calibrated?

- Creating a temperature scale requires 2 fixed points
- The fixed points need to be reliable so all thermometers match
- Scale can then be made by dividing the space between the fixed points

How could you identify fixed points for a thermometer being used at home?

- Freezing and boiling point of water are relatively easy to measure
- Place a thermometer in pure, melting ice and mark where it reads
- Place the same thermometer in a sealed container of steam and mark the scale
- This gives 0 and 100 °C, the scale between can now be divided up

What is meant by the sensitivity, range and linearity of a thermometer?

- *Sensitivity: how small a change in temperature it can detect*
- *Range: the width between min and max temperature*
- *Linearity: how constant the changes measured are as temperature varies*

How does a liquid-in-glass thermometer work?

- Normally filled with mercury or alcohol
- Uses the expansion and contraction of liquids with changes in temperature
- As volume changes they fill or empty the glass tube marked with a scale

How will the structure of a liquid-in-gas thermometer relate to its sensitivity, range and linearity?

- *Width of the tube will affect sensitivity —> narrower tubes result in larger changes against the scale*
- *Type of liquid will affect range —> different liquids have different freezing and boiling points*
- *Amount of expansion varies slightly (differing linearity) for different temperatures*
 - *This will be different for different liquids*
 - *Thermometers containing different liquids can vary slightly between fixed points*

How does a thermocouple thermometer work? Why would you use one of these?

- Temperature difference between probe and junction creates small voltage
- Voltage is proportional to temperature difference
- Voltage causes a current to flow which can be measured
- Good for large temperature ranges and high temperatures
- Good for rapid reading so useful when temperature varies rapidly

What is the equation for thermal heat capacity?

- Thermal capacity = mass (m) x specific heat capacity (c)

What is meant by specific heat capacity?

- The amount of energy needed to heat 1kg of material by 1°C

How could you find the specific heat capacity of a substance?

- Equipment: beaker with 0.5kg of water, 100W heater, thermometer
- Method:
 - Enclose the apparatus in insulated container
 - Submerge the heater in the water and turn it on for 230 seconds
 - Measure the temperature change
- Results: water heats up by 10°C
 - Change in energy = mass(m) x specific heat capacity(c) x change in temperature(ΔT)
 - Change in energy = 100W x 230s = 23,000J
 - $\Delta T = 10^\circ\text{C}$
 - $m = 0.5\text{kg}$
 - Therefore $c = 4600 \text{ J/kg}^\circ\text{C}$

Explain the relationship between temperature of an object and its internal energy

- Temperature measures how hot something is, measured in Celsius or Kelvin
- Higher temperatures means more internal energy; heat measured in Joules
- Thermal energy makes the molecules of a body vibrate more
- This vibration means each molecule is carrying more energy

What is the equation relating energy to thermal heat capacity?

- Change in energy = $mc\Delta T$

E.g. 1kg of material 'A' requires 20,000J to raise its temperature 1°C. 1kg material 'B' requires 10,000J for a 1°C rise. What can be said about material 'A'?

- Thermal capacity is a ratio of energy change in a body to the temperature change
- Material 'A' needs to absorb twice as much energy to raise its temperature
- So material 'A' has a higher thermal capacity

Describe the process of boiling

- Regions within a liquid turning to gas
- Gas rises through the liquid releasing vapour from the surface
- At boiling point - temperature stays the same despite continued energy input
- Continued energy input separates the molecules to form gas

Describe the process of melting

- Solid turns to liquid
- Thermal energy is absorbed $\rightarrow$ temperature rises
- Energy eventually sufficient to overcome molecular attraction
- At this point more energy does not increase temperature, but separates the molecules faster

What does 'melting point' mean?

- The temperature at which a solid will turn to liquid
- Different for each element or compound

What is a boiling point?

- Temperature at which a liquid will turn to gas

What is condensation?

- Gas turns to liquid
- Gas particles lose thermal energy and move closer together

What is solidification?

- Liquid turns to solid
- Liquid loses thermal energy and molecules move closer together

What is the difference between boiling and evaporation?

- *Boiling:*
 - o Rapid
 - o Vapour bubbles in liquid have high kinetic energy
 - o Bubbles rise to top of liquid and burst
- *Evaporation:*
 - o Slower
 - o Fewer molecules have enough energy to turn to gas
 - o Takes place at surface of liquid

What is latent heat of fusion?

- Energy absorbed by a solid as it melts
- Energy needed to separate particles
- Temperature of the solid does not change

What is latent heat of vaporisation?

- Energy absorbed by a boiling liquid
- Does not change the liquid's temperature
- Needed to separate particles so they can form a gas

What is specific latent heat?

- The energy needed to change each kg of solid to liquid (or liquid to gas)
- Given in J/kg
- $\text{Energy} = \text{mass (m)} \times \text{specific latent heat (l)}$
- Example: how much energy is needed to change 5kg of ice into water?
 - o Latent heat of fusion of ice = 330000 J/kg
 - o Energy transferred = 5×330000
 - o = 1650000 Joules

How would you measure specific latent heat for steam?

- Beaker of water on a balance to measure mass
- Heat water using a 100W heater for 500 seconds
- Calculate energy supplied using $\text{power (W)} = \text{energy (J)} / \text{time (s)}$
- Rearrange to make $\text{energy (J)} = \text{power (W)} \times \text{time (s)}$
- Measure mass of liquid turned to gas (reduction in mass)
- Calculate specific latent heat using;
 - o $\text{Specific latent heat} = \text{energy transferred} / \text{mass}$

How would you measure specific latent heat of ice?

- Funnel filled with ice, an electric heater of known wattage, heat for measured time
- Beaker to collect water
- Calculate energy transferred = $\text{power (W)} \times \text{time (s)}$
- Measure mass of ice melted
- Calculate specific latent heat by:
 - o $\text{Specific latent heat} = \text{energy transferred} / \text{mass}$

2.3 Thermal Processes

Describe an experiment to investigate properties of good thermal conductors

- Place rods of different types of metal in boiling water
- Coat exposed ends with thin layer of wax
- Measure amount of wax melted after a given time

Describe an experiment to investigate properties of poor thermal conductors

- Place ice and water in test tube
- Trap ice at bottom of test tube with gauze
- Heat water at top with a Bunsen burner
- Water at top of tube can boil without ice melting – shows water is a poor conductor

What is conduction?

- *Material is heated, particles vibrate faster in the lattice*
- *Causes neighbouring particles to vibrate faster – energy is transferred*
- *Free electrons also speed up when heated*
- *Electrons collide with atoms causing them to vibrate faster*
- *Thermal energy distributed throughout material*

What is convection?

- Method of heat transfer in fluids
- As fluid is heated it becomes less dense —> rises
- Denser water/gas replaces it and is heated, causing a current
- Only occurs when fluid heated at the bottom, not at the top

How can convection be illustrated in an experiment?

- Water in a large beaker with potassium permanganate in one corner
- Heat water in that corner
- Observe coloured water circulate by convection

What is infrared radiation?

- Part of electromagnetic spectrum
- Also called thermal radiation
- Heats up objects when absorbed
- Can travel across a vacuum

Effects on radiation of colour and texture

	Matt black	White	Shiny silver
Emitting	Best	Poor	Worst
Absorbing	Best	Poor	Worst
Reflecting	Worst	Good	Best

How can these properties be demonstrated in an experiment?

Comparing emitters:

- *Place boiling water in a metal cube with different colours/textures each side*
- *Measure thermal radiation emitted from each side using a meter*
- *Compare which surface emits most*

Comparing absorbers:

- *Place a radiant heater at equal distances from metal plates of different colour/ texture*
- *Measure temperature of plates to compare which absorbs most*

What factors affect radiation emitted?

- Surface temperature
- Surface area

Give examples of everyday applications of conduction, convection and radiation

Conduction	Convection	Radiation
Cooker hob	Heating water in a saucepan	Central heating
Frying pan	Fan heater	Sunlight

3. Properties of Waves, Including Light and Sound

3.1 General Wave Properties

What is a wave?

- A wave can transfer energy without transferring matter

What is wave motion?

- Transverse or longitudinal
- Transverse waves move at right angles to direction of travel
- Longitudinal waves move in same direction as travel
- Can be demonstrated by vibrations in a spring or with water waves

Transverse waves:

- Oscillations move at right angles to direction of travel
- Examples are electromagnetic waves

Longitudinal waves:

- Composed of compressions and rarefactions
- Examples are sound waves

What is a wavefront?

- The peaks of transverse waves
- The compressions of a longitudinal wave
- Used to illustrate wave motion

Wave descriptors:

- Speed – how fast the waves move in m/s
- Frequency – number of waves passing a point per second
- Wavelength – distance between peaks
- Amplitude – maximum distance a point moves from 0 as wave passes

What is the wave equation?

- $\text{Speed} = \text{frequency} \times \text{wavelength}$
- $v = f\lambda$

Example: A wave travels at 8 m/s with 2m wavelength, what is the frequency?

- $f = v/\lambda$
- $f = 8/2 = 4 \text{ Hz}$

What is reflection?

- Waves hit a vertical/plane surface
- Reflected from the surface at the same angle as they strike it

What is refraction?

- Waves are slowed by an obstacle of a different medium (e.g. water, glass)
- The reduction in speed changes the direction of travel

What is diffraction?

- Waves travel through a narrow gap
- Waves bend around the side
- Waves spread out as they pass through the gap
- *Only significant if gap size is about same as wavelength*
- *Wider gaps cause less diffraction*
- *Longer wavelengths are more diffracted by an edge than shorter wavelengths*

3.2 Light

How is an optical image formed in a mirror?

- Light from an object strikes a plane mirror and is reflected from mirror surface
- After reflection, light strikes the eye of the observer
- Image appears in the mirror laterally inverted (back to front)
- Image is same size as object
- Image is as far behind mirror as object is in front
- A line through equivalent points of image and object passes through mirror at right angles
- *Image in a plane mirror is virtual*

What is a normal line?

- A line drawn perpendicular to the mirror

What is the law of reflection?

- Angle of incidence = angle of reflection

Working out image position

- *Draw a line from the object through the mirror at 90 degrees, extending well behind the mirror*
- *Measure the distance from the object to the mirror*
- *At an equal distance behind the mirror on the line is the image position*

What is refraction?

- A change in direction of waves when they travel across a boundary from one medium to another

Describe how a light ray changes direction when it enters and leaves a glass block

- *Bends towards the normal as it enters the block*
- *Bends away from the normal as it leaves the block*

Give an experimental demonstration of light refraction

- Looking at an object underwater

Describe how light passes through parallel-sided transparent material

- Light strikes the material at angle of incidence i
- Light refracted towards the normal as it enters, at angle of refraction r
- Refracted away from the normal as it leaves
- Ray emerges parallel to original after exit

What are the equations for the refractive index?

- $n = \sin i / \sin r$
- $n = 1/\sin c$

What is the refractive index?

- *How much a medium alters the speed of light*
- *Refractive index (n) = speed of light in vacuum / speed of light in medium*
- *E.g. if $n = 1.5$ for glass, and the speed of light in a vacuum is 300000 km/s*
- *Rearrange for speed of light in medium = speed of light in vacuum / n*
- *Speed of light in glass = 300000 / 1.5 = 200000 km/s*

What is the relationship between angle of incidence and angle of refraction?

- They are always proportional
- $\sin i / \sin r = n$
- Example: if light strikes glass at an angle of 40, the refractive index is 1.5, what is the angle of refraction?
- Rearrange equation so $\sin r = \sin 40 / 1.5$
- $\sin r = 0.4285$
- $r = 25$ degrees

Define critical angle

- Angle of incidence beyond which rays of light are totally internally reflected

Define total internal reflection

- When the angle of incidence is greater than the critical angle
- All light is reflected, none is refracted

How is the critical angle calculated?

- $n = 1 / \sin c$
- example : if $n = 1.3$, what is the critical angle?
- Rearrange to get $\sin c = 1/n$
- $\sin c = 1/1.3 = 0.769$
- $c = 40$ degrees

How do optical fibres work?

- Total internal reflection
- **Core** of fibre made out of glass with **high refractive index**
- **Cladding** is made out of glass with **lower refractive index**
- **Light** entering core is at an angle greater than the critical angle so light is TIR

Give examples of the use of optical fibres

- Endoscope - used in keyhole surgery
- Communication as less energy is lost when compared to using copper wiring

How does a thin converging lens work?

- Converging lens is a convex lens
- Rays parallel to principal axis are bent as they pass through
- Point where rays meet is principal focus
- Forms real images on screen
- Focal length is distance between lens and principal focus

Images produced by converging lenses are:

- Real (can be displayed on a screen)
- Inverted (upside down)
- Can be enlarged if object is near the focal length, smaller if further away

How can a single lens be used as a magnifying glass?

- Place object closer to convex lens than the principal focus
- Rays never converge (image is virtual)
- Image is upright and magnified

What is the difference between a real and a virtual image?

- Real image can be displayed on a screen
- Rays forming a virtual image never converge, can't be displayed on screen

How does a prism disperse light?

- Prism sides not parallel, light refracted
- Prism refracts each colour by a different amount
- Light splits into range of colours (spectrum)
- Order = Red, orange, yellow, green, blue, indigo, violet

What is the name for light of a single frequency?

- *Monochromatic*

3.3 Electromagnetic Spectrum

What are the main features of electromagnetic waves?

- Transfer energy without transferring matter
- Transverse waves - oscillations at right angles to direction of travel
- Travel with the same speed through a vacuum as in air
- Speed is 3.0×10^8 m/s

Type	Frequency	Wavelength	Properties & uses (and dangers)
Radio	$10^5 - 10^8$ Hz	$10^0 - 10^{-3}$	• Radio, TV and communications
Microwaves	$10^9 - 10^{11}$ Hz	$10^0 - 10^{-3}$	• Mobile phones • Satellite TV • Telephones
Infrared	$10^{12} - 10^{14}$ Hz	$10^{-3} - 10^{-6}$	• Radiant heaters and grills • Remote controls • Intruder alarms
Light	10^{15}	10^{-6}	• Visible light
Ultraviolet	$10^{16} - 10^{17}$ Hz	$10^{-7} - 10^{-9}$	• Causes tanning, skin cancer, eye damage • Kills bacteria
X-rays	$10^{18} - 10^{19}$ Hz	$10^{-10} - 10^{-11}$	• X ray photography • Causes cancer • Kills cancer cells
Gamma rays	$10^{20} - 10^{22}$ Hz	$10^{-12} - 10^{-14}$	• Emitted by radioactive materials • Sterilising medical equipment • Causes cancer • Kills cancer cells

What are the safety issues with microwaves?

- Heat up water molecules
- Can heat up body tissues internally

What are safety issues with X rays and gamma rays?

- Damage living cells in the body
- Can cause cancer

3.4 Sound

How is sound produced?

- Sources vibrate and produce longitudinal waves
- Waves made up of compressions and rarefactions
- Creates sound
- Needs a medium to be transmitted (can't travel in a vacuum)

Describe compression

- *Regions of air where air particles are compressed together*

Describe rarefactions

- *Lower pressure between compressions*

What is the range of audible frequencies for human ears?

- 20Hz to 20000 Hz

What is ultrasound?

- Sounds above human hearing
- Can be used medically to measure blood flow through heart or kidney

How can speed of sound be determined in an experiment?

- Create sound (e.g. hammer on block)
- Place two microphones at different distances from source
- Measure time lag between detection at microphones
- $\text{Speed of sound} = \text{distance travelled} / \text{time taken}$

Why can't sound travel across a vacuum?

- Requires a medium (e.g. air, water) to be transmitted

How do loudness and pitch change?

- Louder sound = increased amplitude
- Higher pitch = higher frequency (Hz)

What produces an echo?

- Reflection of a sound wave from a surface

Speed of sound:

- *Through air $\rightarrow 330 \text{ m/s}$*
- *Through water at 0 C $\rightarrow 1400 \text{ m/s}$*
- *Through concrete $\rightarrow 5000 \text{ m/s}$*

4. Electricity and Magnetism

4.1 Simple Phenomena of Magnetism

Describe the forces between magnets

- Magnets attract between NORTH and SOUTH poles
- Magnets repel when together (i.e. N-N or S-S)
- Magnets attract a magnetic material to either pole

What are the differences between a magnet and a magnetic material?

Magnet	Magnetic material
• Has a magnetic field • Has 2 opposite poles • Will attract magnetic materials	• Doesn't have a magnetic field • Attracted by a magnet • Can have a magnetic field induced • Hard materials keep magnetic field • Soft materials lose magnetic field • E.g. iron, nickel or cobalt

What is meant by 'induced magnetism'?

- When a non-magnetic material develops magnetism
- Atoms in magnetic material have small magnetic force
- When these are pulled into line the material becomes a magnet

How may an objects' magnetism be induced (created)?

- Place a magnetic object (e.g. one made from steel or iron) into a magnetic field
- Object becomes a magnet (magnetism has been induced)
- The magnetism is temporary as the object loses its magnetism when it's removed from the magnetic field

How can a material become magnetised?

- Happens with weak effect when held next to a magnet
- Stroking with one pole of a magnet has a stronger effect
- Happens strongly when inside a wire coil with directional current
- Hammering a magnetic material while in a magnetic field

What will cause a loss of magnetism?

- *Hammering causes the atoms to become disorganised*
- *Heating to a high temperature causes atoms to become disorganised*
- *Placing magnet in a coil and passing an alternating current through it*

Define magnetic field line

- The space around a magnet where magnetism can be detected
- Magnetic field line can be seen through use of either plotting compasses or iron filings

How could you find the pattern of field lines and direction?

- Use small compass to plot the line
- Start near one end of magnet and mark the direction the needle points
- Move the compass to a new position a little distance away
- When the compass lines up with the previous dot mark the new position
- Joining all the dots from pole to pole
- The direction of the compass needle gives direction

What is the difference between steel and iron's magnetism?

- Steel - hard magnetic material - retains magnetism
- Iron - soft magnetic material - loses magnetism

What are the differences in design of a permanent magnet and electromagnet?

- Electromagnets → soft magnetic core surrounded by coils of wire
- Can be turned on (magnetic) and off (non-magnetic)
- Used for switches, circuit breakers etc.
- Permanent magnets → hard magnetic materials
- Always magnetic
- Used for compasses, in speakers etc.

4.2 Electrical Quantities

What are the two possible electrical charges?

- Positive
- Negative

How do electrical charges interact?

- Opposite charges attract
- Like charges repel

How does a body become charged?

- Addition or removal of electrons

What is the unit for charge?

- *Coulombs*

Describe experiments to show production of electric charge

- Rubbing polythene with a woollen cloth
- Rubbing perspex with a woollen cloth

Describe an experiment to detect charge

- Use a leaf electroscope to detect charge
- Place charged object near the electroscope

Give examples of electrical conductors and insulators

Conductors	Semi conductors	Insulators
Metals e.g. silver, copper	Silicon	Plastics e.g. PVC, polythene
Carbon	Germanium	Glass
Water (poor conductor)		Rubber
Human body (poor)		Dry air

Describe the direction of an electric field

- *The direction of the force on a positive charge at that point (e.g. towards the negative terminal of a battery)*

What is an electric field?

- *Region in which electric charge experiences a force*

Describe electric field patterns

- *Lines with arrows represent fields*
- *Arrows show where force on a positive charge would act*
- *Field lines point away from positive, towards negative*

What is charging by induction?

- *Charging an uncharged object by proximity*
- *Object never touches – electrons are pulled near or repelled from charge*
- *Results in opposite charge to the charged object*

What makes good conductors carry electricity?

- *Free electrons that can move through a material easily*
- *E.g. metals with loosely held outer electrons*

What makes an insulator not conduct?

- *Tightly held electrons*
- *Not free to move*

What is current?

- *Rate of flow of charge*
- *In metals → flow of electrons*
- *Charge = current $\times$ time*
- *$Q = I \times t$*

Example: if a current of 3 amperes flows for 5 seconds, what is the charge delivered?

- *$I = 3, t = 5$*
- *$Q = 3 \times 5$*
- *$Q = 15 \text{ Coulombs}$*

What is an ammeter?

- *Measures current in amperes*
- *Connected in series*
- *Can be analogue or digital*

What is conventional current and how does it relate to flow of electrons?

- *Conventional current is depicted as flowing from positive to negative*
- *Used for historical reasons*
- *Electrons flow the opposite way*

What is electromotive force (e.m.f)?

- *A force measured in volts*
- *For electrical sources of energy*
- *Work done per unit of charge by the cell in driving charge round the complete circuit*
- *Maximum potential difference when not in circuit*
- *Drops when current being supplied due to energy wastage in cell*

What is potential difference?

- *Force measured in volts*
- *Measured by a voltmeter*
- *Describes energy given to electrons pushed out*
- *Defines how much energy given to each coulomb of charge*

How is a voltmeter used?

- *Connected in parallel*
- *Reads voltage across the component*
- *Can be analogue or digital*
- *1 V is equivalent to 1 J/C*

What is resistance?

- How easily current flows in a circuit/component/material
- Measured in Ohms Ω
- Resistance = potential difference (PD) / current
- $R = V / I$
- Increased resistance reduces current

Example: if a voltage of 4V is needed to make a 2A current, what is the resistance?

- $R = V / I$
- $R = 4/2$
- $R = 2 \Omega$

What is Ohm's law?

- The current through a resistor at constant temperature is directly proportional to the potential difference across the resistor

Describe an experiment to find resistance using a voltmeter and ammeter

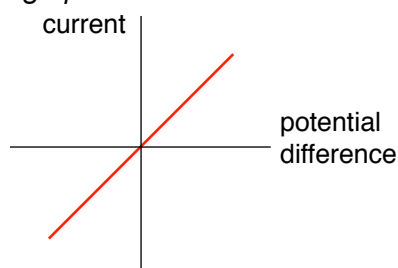
- Connect an ammeter in a circuit and a voltmeter across a component
- Measure voltage and current and use $R = V/I$

How does resistance of a wire relate to its length and diameter?

- Doubling length $\rightarrow$ doubles resistance
- *Resistance is directly proportional to length*
- Halving cross sectional area $\rightarrow$ doubles resistance
- *Resistance is inversely proportional to cross sectional diameter*
- $R = \rho \times (1/A)$ – where ρ is the resistivity, a constant for the material
- If ρ is the same for two materials, then:
 - o $\text{Resistance}_A \times \text{area}_A / \text{length}_A = \text{resistance}_B \times \text{area}_B / \text{length}_B$

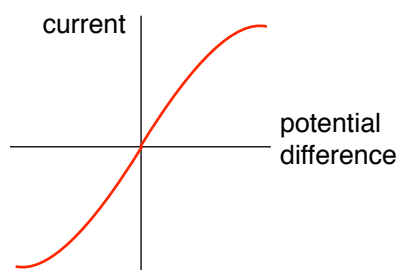
Sketch and explain the current-voltage characteristic of an ohmic resistor

- *Resistance is always constant*
- *Linear graph where current is directly proportional to voltage*



Sketch and explain the current-voltage characteristic of a filament lamp

- *As current increases, temperature rises and resistance goes up*
- *Current is not proportional to the voltage*

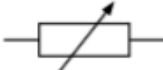
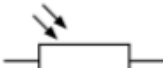


How do electric circuits transfer energy?

- Energy from the cell or power source transfers to circuit components, then to surroundings
- *Power = voltage x current ($P=V \times I$)*
 - o *As energy transferred = $P \times t$*
 - o *Energy transferred = voltage x current x time ($E=V \times I \times t$)*

4.3 Electric Circuits

What do the following symbols in a circuit diagram represent?

cell		switch	
battery of cells		earth or ground	
or			
power supply		electric bell	
a.c. power supply		buzzer	
junction of conductors		microphone	
lamp		loudspeaker	
fixed resistor		motor	
variable resistor		generator	
thermistor		ammeter	
light dependent resistor		voltmeter	
heater		galvanometer	
potential divider		diode	
relay coil		light-emitting diode	
transformer		fuse	

Symbols courtesy of CIE IGCSE (9-1) Physics Syllabus 0972

What are the current, voltage and resistance rules in a series circuit?

- Current is the same everywhere
- Total voltage is the sum of all the individual components' voltage
- Total resistance is the sum of all the individual components' resistance

What are the current, voltage and resistance rules in a parallel circuit?

- Voltage is the same everywhere
- Total current is the sum of the individual components' current
- Combined resistance of two resistors in parallel is less than that of either resistor by itself
- *Effective resistance = Product of resistors / Sum of resistors*

Why is current conserved at a junction?

- In parallel circuit, current is shared between each component
- Total amount of current flowing into the junction is equal to the total current flowing out
- Current is described as being conserved

If multiple power sources are in series what is their total electromotive force (e.m.f)?

- *Total e.m.f in series is the sum of the individual sources*

If the sum of the PD across the components of a series circuit is 12v what is the PD of the source?

- *PD of components = PD of the source*
- *PD of the source is 12v*

What is the advantage of connecting lamps in parallel?

- Each bulb gets the same P.D despite differing current
- Both lamps glow brightly

What does a variable potential divider (potentiometer) do?

- Uses a variable resistor to take a portion of the source's voltage
- This voltage can be delivered to a separate circuit
- The amount delivered can be varied by varying the resistor

What is a rectifier?

- *A component that can change an alternating current into a direct current*

Describe the action of a diode in a circuit

- *Diodes allow current to flow in one direction only*
- *Used to produce direct current from an alternating current source*
- *Used this way they act as a rectifier*

How does a relay work as a switch for a circuit?

- A switch on a large powerful circuit operated by a separate small circuit
- Electromagnet switched on in first circuit
- Attracts the contact of the second circuit, closes switch

Describe how a light-dependent resistor works. How can it be used?

- Resistance decreases as light intensity increases
- Used in circuits as an input transducer
- *Used as part of a light sensitive switch*
- *Placed in a potential divider to deliver voltage to a lamp*
- *Lamp will come on when it is dark*

Describe how a thermistor works. How can it be used?

- Resistance decreases as temperature increases
- Used as an input transducer in circuits needed to be sensitive to temperature
- *E.g. in a fire alarm*
- *Placed in a potential divider to deliver high P.D in high temperatures*
- *Attached to a relay, high P.D used to turn on an electromagnet*
- *Electromagnet turns on a switch and the alarm rings*

4.4 Digital Electronics

Explain the term 'analogue'

- Continuously variable signal based on input

Explain 'digital'

- Signal is either on (high / 1) or off (low / 0) with no variation

How would you draw logic gates in a circuit diagram?

AND gate



NOR gate



OR gate



NOT gate



NAND gate



Symbols courtesy of CIE IGCSE (9-1) Physics Syllabus 0972

Describe the actions of the following logic gates:

Logic Gate	Action
AND	- Has 2 inputs, 1 output - If both input 1 AND 2 is high, then output is high
OR	- Has 2 inputs, 1 output - If input 1 OR 2 is high then output is high
NOT	1 input, 1 output - If input is high then output is NOT high - If input is low then output is NOT low
NAND	2 inputs, 1 output - An AND gate inverted by a NOT gate - If input 1 AND 2 are high then output are NOT high
NOR	2 inputs, 1 output - An OR gate inverted by a NOT gate - If neither input 1 NOR input 2 are high then output is high

Complete a truth table for an AND gate

Inputs		Output
A	B	Q
0	0	0
0	1	0
1	0	0
1	1	1

Complete a truth table for an OR gate

Inputs		Output
A	B	Q
0	0	0
0	1	1
1	0	1
1	1	1

Complete a truth table for a NOT gate

Inputs	Output
A	Q
0	1
1	0

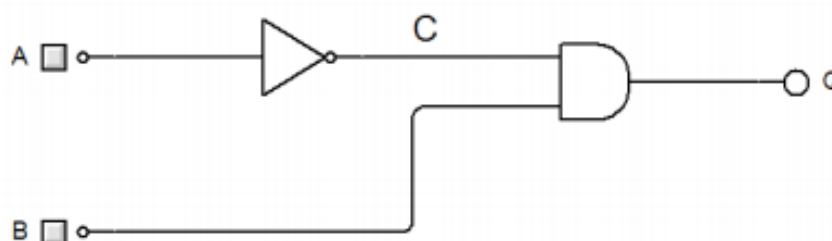
Complete a truth table for a NAND gate

Inputs		Output
A	B	Q
0	0	1
0	1	1
1	0	1
1	1	0

Complete a truth table for a NOR gate

Inputs		Output
A	B	Q
0	0	1
0	1	0
1	0	0
1	1	0

Example: Complete the truth table for the following logic system



Inputs		Outputs	
A	B	C	Q
0	0	1	0
0	1	1	1
1	0	0	0
1	1	0	0

4.5 Dangers of Electricity

What are the hazards of damaged insulation, overheating of cables and damp conditions?

- Electric shock from exposed wiring or water in sockets or plugs
- Electrical fires from overheated cables or frayed wiring

What is the function of a fuse?

- Protect the circuit and user of appliance
- Overheats and melts if current too high
- Breaks circuit before wires/components get too hot and catch fire

How does a circuit breaker work?

- An automatic switch that 'trips' (turns off) when current too high
- Can be reset

How do you choose the correct fuse or circuit breaker setting for a device?

- Rating must be above normal current for the device
- As close to normal current as possible so circuit cannot overheat

Example: Would a 10A fuse be suitable for a 1600W hairdryer with a PD of 200V?

- Current = power / voltage
- $1600 / 200 = 8 \text{ A}$
- 10A fuse would be suitable

Why should metal cases be earthed?

- Prevents electrocution if live wire comes loose and touches case
- Earth wire carries current to earth and blows the fuse

What is double insulation and why is it used?

- Both the wires inside a device and the outer case of the device are insulated
- No chance of electrocution

4.6 Electromagnetic Effects

What is electromagnetic induction?

- Magnetic field is used to create a current
- A conductor (e.g. a wire) is moved across a magnetic field (or a changing magnetic field moves around a conductor)
- Electromotive force (EMF) is induced in the conductor
- If a complete circuit, current flows

Describe an experiment to demonstrate electromagnetic induction

- Connect a coil in a circuit with a galvanometer to measure current
- Move a magnet through the coil
- Record the current induced

What factors affect magnitude of induced EMF?

- Speed of wire movement
- Strength of magnet
- Length of wire in the magnetic field

What dictates direction of the induced EMF?

- *It opposes the change causing it*
- *E.g. magnet moving towards a coil - induced current turns coil into magnet that repels approaching magnet*

How do you tell which direction the current will flow when moving a wire through a magnetic field?

- *Right hand rule applies*
- *Make 'L-shape' with thumb and index finger of right hand*
- *Point middle finger perpendicular to thumb and index finger*
- *Index finger → direction current is flowing*
- *Middle finger → direction of the magnetic field*
- *Thumb → direction the wire is pushed*

Which direction is the force produced when a current is within a magnetic field?

- *Left hand rule applies*
- *Make 'L-shape' with thumb and index finger of left hand*
- *Point middle finger perpendicular to thumb and index finger*
- *Index finger → magnetic field*
- *Middle finger → direction of current*
- *Thumb → direction of force*

How do you tell the poles of an induced electromagnet?

- *Right hand grip rule applies*
- *Make 'thumbs up' gesture with right hand*
- *Curl fingers in same direction as the magnetic field around the wire*
- *Thumb will point in direction of current flow*

What is the difference between a/c and d/c

- a/c is alternating current, flows alternately backwards and forwards
 - E.g. mains electricity
- d/c is direct current, flows one way only
 - E.g. a battery

How does a rotating coil generator work?

- *Coil of copper wire, rotated in magnetic field*
- *Produces alternating current as it rotates*
- *Slip rings are contacts and rotate with the coil*
- *Carbon brushes take current from slip rings in to circuit*

Relate the position of generator coil to peaks and zeros of voltage output

- Peaks in p.d. correlate to coil at 90 degrees to magnetic field
- Zero points relate to coil parallel to magnetic field

Describe construction of a basic transformer with a soft-iron core

- Electromagnet connected to a/c source
- Placed next to a second electromagnet
- a/c in first electromagnet creates a changing magnetic field
- Changing magnetic field induces current in next electromagnet

How do input voltage and output voltage relate?

- Output voltage / Input voltage = No. of turns on output coil / no. of turns on input coil
- $V_2/V_1 = N_2/N_1$

Example: if the input is 10V and 1000 coils, and the output is 2000 coils, what will be the output voltage?

- Output voltage = (turns on output coil / turns on input coil) x input voltage
- Output voltage = (2000/1000) x 10 = 20 V

What does step-up mean?

- Using a transformer to increase the voltage

What does step-down mean?

- Using a transformer to decrease the voltage

How is a transformer used in high voltage electricity transmission?

- Step-up transformer produces high voltage and low current for long distances
- Step-down used to reduce voltage but increase current for usage

What is the advantage of high voltage transmission?

- Less power is lost due to resistance of wires
- $\text{Current}^2 \times \text{resistance} = \text{power lost}$
- By decreasing current and increasing voltage, less power lost

How do step-up and step-down transformers differ from each other?

- Step-up transformers have more turns on the secondary coil than primary
 - o Increase size of alternating potential difference
- Step-down transformers have fewer turns on the secondary coil than the primary
 - o Decrease size of alternating potential difference

How is efficiency calculated?

- Input voltage x input current = output voltage x output current
- $V_1 I_1 = V_2 I_2$

Example: If 10V at 2 A flows into a step-up transformer, which produces an output voltage of 20V, what is the output current?

- $V_1 I_1 = V_2 I_2$
- $I_2 = (10 \times 2) / 20$
- = 1A

What is a solenoid?

- Coil of wire

How can the magnetic field around a current carrying wire be made stronger?

- Increasing current
- Wrapping the wire into a coil (solenoid)

How can the magnetic field around a solenoid be made stronger?

- Increasing the number of turns on the solenoid
- Increasing the current
- Adding a soft iron core

What is the pattern of a magnetic field around a straight wire?

- It is perpendicular to the wire at all points
- Circular
- Right hand grip rule gives direction of field
- *Strongest closest to the wire*

What is the pattern of a magnetic field around a solenoid?

- Same pattern as a bar magnet
- Magnetic poles at the ends of the coil
- *Right-hand grip rule gives direction of N pole*

What is the effect on the magnetic field of increasing the magnitude of current?

- *Increased current = increased strength of magnetic field*

What is the effect on the magnetic field of changing the direction of current?

- *Change in direction of current will reverse the magnetic field*

Describe an experiment to show the force on a conductor in a magnetic field:

- Place a wire between N and S poles on a magnet
- Run a direct current through the wire
- Observe movement
- Reversing current causes opposite movement direction
- Opposite magnetic field direction gives opposite movement
- *The resultant force is denoted by the left hand rule*

Describe an experiment to show magnetic force on beams of charged particles

- *Use an electron gun*
- *Fires beam of electrons across a fluorescent screen, shows path of beam*
- *Put a magnetic field perpendicular to screen*
- *Observe movement of path of beam on screen*

How does a d/c motor work?

- A current flows through the wire creating a magnetic field around the wire
- This temporary magnetic field interacts with the permanent magnetic field of the bar magnets
- A force is created
- The force turns the coil of wire
- *As each side passes vertical, the current is reversed to maintain turning in same direction*
- *This is achieved by a split ring commutator in a d/c motor*

How is the turning effect of a motor increased?

- Increasing number of turns on the coil
- Increasing current
- Increasing strength of the magnetic field

5. Atomic Physics

5.1 The Nuclear Atom

What is the structure of an atom?

- Positively charged nucleus
 - Contains protons and neutrons
- Negatively charged electrons orbit nucleus

	Mass	Charge
Electron	1/2000	-1
Proton	1	+1
Neutron	1	0

What is the evidence for the nuclear atom?

- Rutherford's gold foil experiment
- Positive alpha particles fired at thin gold film
- Most pass through, some are deflected
- Shows that majority of atom is empty space, nucleus is positively charged

What is the composition of the nucleus?

- Made up of protons and neutrons
- Protons are positively charged (relative charge +1)
- Neutrons are neutral (relative charge 0)

What is the nucleon number (A)?

- Combined number of protons and neutrons in nucleus
- Also known as mass number

What is the proton number?

- Number of protons in the nucleus
- Always the same for an element

Example:



- Na = sodium
- Mass/nucleon number = 23
- Atomic number = 11
- Proton number = 11
- Neutron number = 12

What is a nuclide?

- A specific variation of an element X where the nucleon number varies
- Denoted by the notation ^A_ZX where A = nucleon number, Z = proton number

What is an isotope?

- An atom of the same element with the same number of protons but different number of neutrons
- Nucleon number varies but proton number is the same

What is nuclear fission?

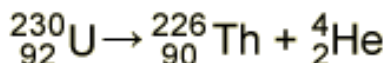
- Splitting a nucleus
- Produce energy, two daughter nuclei and releases 2-3 neutrons

What is nuclear fusion?

- The creation of larger nuclei resulting in the loss of mass from small nuclei
- Releases energy

How is radioactive decay written in nuclide notation?

- Top and bottom numbers must balance
- E.g.



5.2 Radioactivity

What is background radiation?

- Radiation which is always present
- Produced by rocks, soil, cosmic rays, radon gas

What are α-particles?

- A helium nucleus
- 2 protons and 2 neutrons
- Mass number decreases by 4, atomic number (proton number) decreases by 2

How are alpha (α) particles detected?

- In a cloud chamber
- Contains cold alcohol vapour in air
- Alpha particles moving through cloud make visible trails of condensed alcohol
- Also detected by a Geiger-Muller tube

What are beta (β) particles?

- Fast moving electrons
- Emitted when neutron splits into a proton and electron
- Mass number unchanged, atomic number (proton number) increases by 1

What are gamma (γ) rays?

- Electromagnetic waves
- Emitted after an α or β particle has been emitted
- No change to mass or atomic number

How are β particles and γ rays detected?

- With a Geiger-Muller tube
- Use a paper screen to block out alpha particles
- Aluminium foil will block beta particles
- Tube detects current as radiation ionises gas held inside tube

What is the nature of radioactive emission?

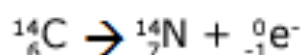
- Happens spontaneously and at random
- Cannot predict decay or direction
- Not affected by temperature, pressure or chemical change

Compare the properties of alpha, beta and gamma radiation

	Alpha	Beta	Gamma
Nature	- Two protons and two neutrons (Helium nucleus) - Large - Slow moving - Positively charged (+2)	- Electrons - Small - Fast - Negative charge (-1)	- Electromagnetic wave - Moves at speed of light - No charge
Ionising effect	- Strong - Due to higher charge and larger mass	- Weak - Due to lower charge and small mass	- Very weak - Has no charge or mass
Penetrating ability	- Poor - Stopped by skin, paper, a few cm of air	- Moderate - Stopped by a few mm of metal (e.g. aluminium foil)	- Very penetrating - Never completely stopped - Lead and thick concrete reduce intensity
Examples of use	Smoke detectors	Thickness monitoring	- Radiotherapy - Testing for cracks

What is radioactive decay?

- When unstable nuclei change in to stable ones
- Releasing energy as they do so, in form of radiation
- In alpha or beta decay the nucleus changes to that of a different element
- E.g.



What is half-life?

- Time taken for half the radioactive nuclei to decay
- A random process

What is the effect of ionising radiation on living things?

- Damages living cells
 - Damages DNA structure
 - Causes mutation that can lead to cancer
- Creates ions which can cause further damage

How are radioactive materials handled, used and stored safely?

- Protective clothing worn (e.g. radiation suits)
- Stored behind protective material depending on type of radiation
- Used at a distance
- Checks for contamination before personnel leave site
- Purpose made reactors and equipment to contain radiation
- Wearing radiation monitors to record exposure

What safety measures are used in nuclear power plants? What is the role of the control rods?

- Control rods absorb neutrons to slow down the reaction
- Moderators slow the neutrons
- Lead shielding stops radiation causing cancer in employees